

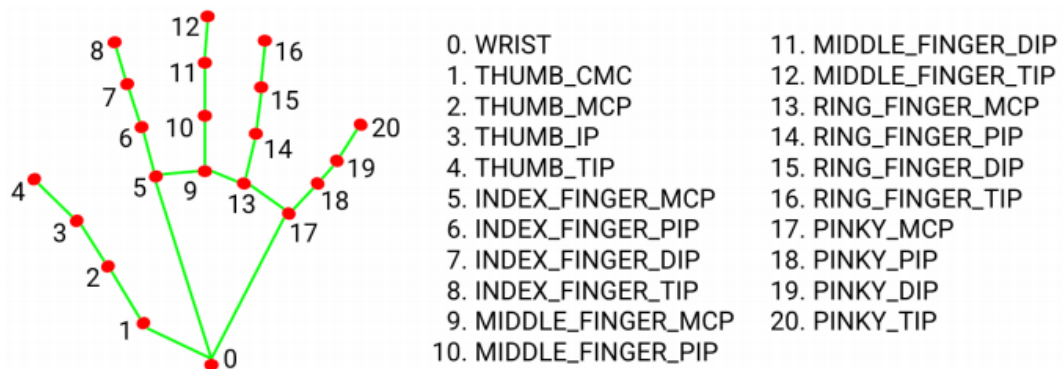
Gesture Control Car Basic Movement

1. Program Function Description

After the function is enabled, the camera captures images and recognizes gestures to control the car's movement.

Gesture	Action
Gesture "5"	Car forward
Gesture "1"	Car turn right
Gesture "2"	Car stop
Gesture "3"	Car backward
Gesture "4"	Car turn left

MediaPipe Hands infers 3D coordinates of 21 hand joint landmarks from a frame



2. Program Startup**

2.1. Startup Commands

For Raspberry Pi 5 users, if the ROS container is not started, you need to start the container first,

```
bringup_m3
```

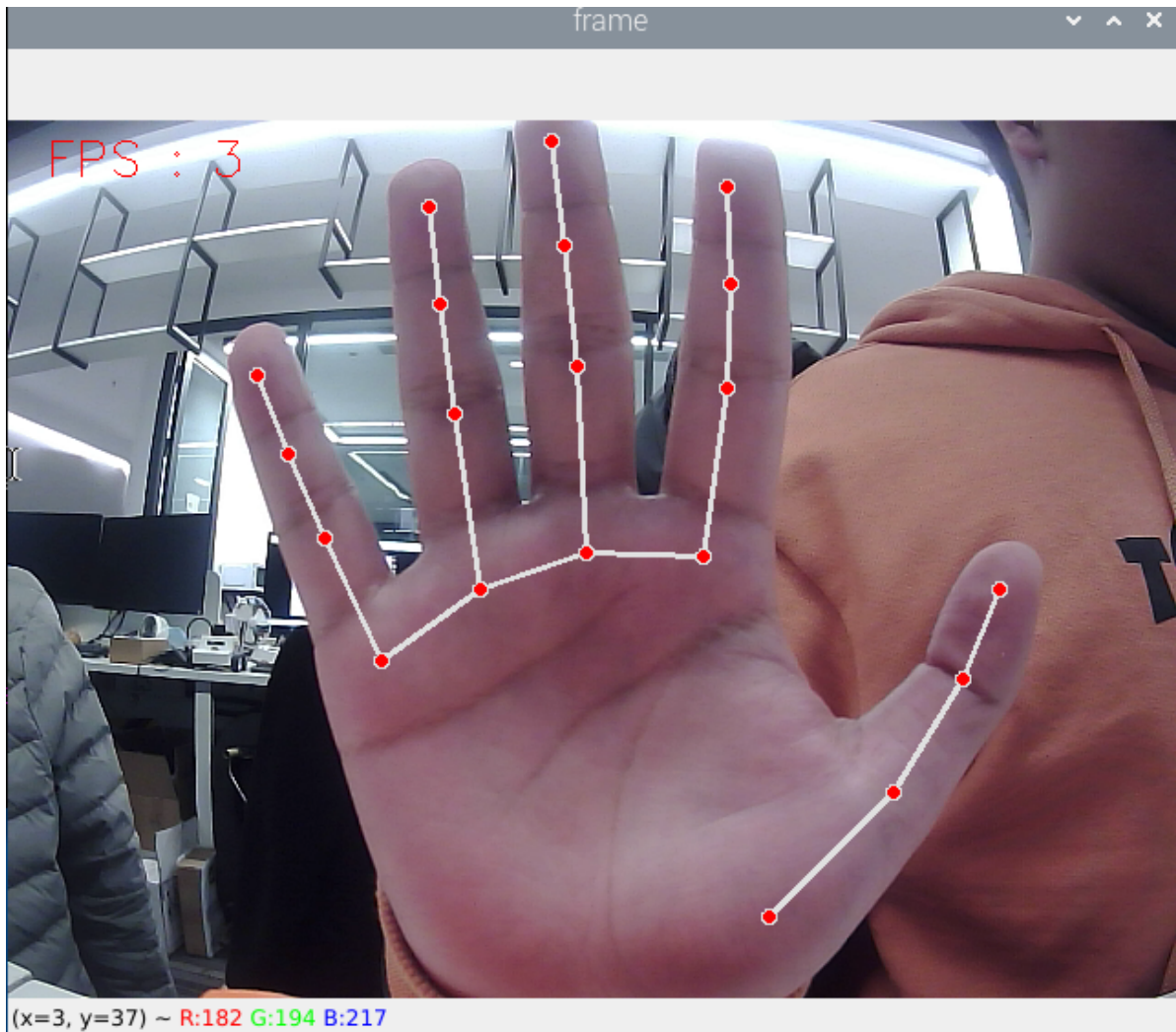
Open a terminal inside the container,

```
exec_m3
```

- For Orin, RDK mainboard, start directly on the car end:

```
ros2 launch m3_bringup mediapipe_demo.launch.py demo:=HandCtrl
```

Enable this function, then place your hand in front of the camera. The screen will draw the finger shapes. After the program recognizes the gesture, it will send the speed to the chassis, thereby controlling the car's movement.



3. Core Code

Program code path:

```
~/yahboomM3_ws/code_ws/src/m3_common_demo/m3_common_demo/mediapipe/14_HandCtrl.py
```

3.1. HandCtrl.py

- Import key libraries

```
from yahboomcar_mediapipe.media_library import * #This library contains functions for hand detection and gesture recognition
```

- Get finger data

```
frame, lmList, _ = self.hand_detector.findHands(frame)
fingers = self.hand_detector.fingersUp(lmList)
sum(fingers)
fingers[]
```

As you can see, first the hand is detected to get the `lmList` value, then it's passed to the `fingersUp` function. The `fingersUp` function is used to detect which fingers are extended, with extended fingers having a value of 1. You can see the specific code in `media_library.py` function, which has detailed explanations. It essentially determines if fingers are extended by judging the `xy` values of finger joints. The `sum(fingers)` function calculates the number of extended fingers, and `fingers[]` can be used to enumerate fingers. For example, the index finger is represented by `fingers[1]`.

- Publish speed to chassis

```
self.media_ros.pub_vel(x,y,z)    #This function is also in media_library.py
```